

SAM3Depth: Depth Discretization with SAM3

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Abstract

Monocular depth estimators such as iDisc decompose a scene through an internal discretization into a small set of internal regions, from which depth is decoded. Whether this discretization carries the depth signal is the question studied here: we investigate whether replacing iDisc’s learned, ungrounded partition with a grounded, object-aware one from a segmentation foundation model (SAM3) improves depth. The probe is a depth head whose discretization source can be swapped between SAM3 decoder queries, projected or refined by attention, and SAM3 object masks. Across both KITTI and NYU, accuracy proves invariant to that source: a linear projection of the queries, an attention refinement, and a mask-grounded variant through the same projection all reach a similar error. We show, however, that the head nonetheless relies on the partition. When the discretization is ablated, error on the SAM3 models grows several-fold, whereas the ResNet-101 baseline is largely unaffected. The partition is therefore used, even though its method of generation and its object-grounding turn out to be irrelevant. Supervision further degrades the grounded structure: under fine-tuning, the region/depth coherence of the masks (measured by R^2) drops from 0.75 to 0.26. This degradation accounts for the failure of object-token tracking to deliver temporal stability in this family of models.

1. Introduction

Monocular depth estimation maps a single RGB image to a dense depth map. Rather than regress every pixel in isolation, many strong models first summarise the scene into a compact decomposition and decode depth relative to it. iDisc [7] makes that decomposition explicit. Its Internal Scene Discretization (ISD) attends a small set of Internal Discrete Representations (IDRs) over the image and reconstructs depth from them, while an Adaptive Feature Partitioning (AFP) module produces the IDRs from the backbone features as learned latent slots without semantic identity.

Since the discretization acts as a bottleneck, a better decomposition might be expected to yield better depth. SAM3, a segmentation foundation model, produces the partition iDisc only learns implicitly, but in grounded form: clean, object-aligned masks together with per-object tokens (Fig. 1). If depth indeed depends on the discretization, sourcing it from SAM3 should help. A grounded and trackable partition was also this project’s original goal, intended to reduce the temporal flicker of frame-by-frame depth by carrying object tokens across frames. The question is therefore whether *the discretization is where the depth signal resides, and whether a better partition gives better depth*. With iDisc’s AFP bottleneck replaced by SAM3 and only the ~ 12 M-parameter head retrained, the discretization becomes straightforward to swap and to ablate.¹

The answer is negative, and the rest of this work makes it precise. The contributions are as follows:

- We show that depth accuracy is invariant to the partition’s source: with one frozen backbone, a linear projection of the queries, an attention refinement, and a mask-grounded source through the same projection all reach nearly-identical AbsRel, on KITTI and again on NYU (Sec. 5.1).
- We run an ablation study on the partition to show that it is used, but independent of the source and grounding (Sec. 5.2).
- We inspect the degradation of the grounded structure under depth supervision: the masks’ region/depth coherence falls from R^2 0.75 to 0.26 under fine-tuning (Secs. 5.3 and 5.4).
- We conclude that, since supervision destroys the object partition, tracking object tokens for temporal stability cannot work in this model family (Sec. 6).

A better or more grounded partition therefore does not seem to produce better depth.

¹Code, configs, and analysis tooling: <https://github.com/Fersoil/idisc>



Figure 1. **Ungrounded vs. grounded partitions.** Left to right: input, iDisc’s AFP partition (fragmented, ungrounded), SAM3 masks (clean, object-grounded). The question is whether the grounded partition forms a better depth bottleneck. Our results indicate that it does not.

2. Related Work

Monocular depth and internal discretization. Modern monocular depth estimators are learned dense regressors. In one family, depth is decoded per pixel from a strong encoder with no explicit scene structure. Such approaches include the transformer decoder DPT and, at scale, Depth Anything [8]. iDisc [7] takes the opposite route and inserts an explicit bottleneck. Its Adaptive Feature Partitioning condenses the image into a handful of Internal Discrete Representations, from which the Internal Scene Discretization decodes depth. That learned, ungrounded partition is the component examined here, and the open question is whether it carries the depth signal or merely reflects a by-product of the features.

Segmentation foundation models. The Segment Anything line [3] produces class-agnostic, promptable object masks. SAM3 [5] extends it with open-vocabulary text grounding and a transformer decoder whose queries correspond to detected objects. Such masks and object queries are precisely the grounded scene partition that iDisc only acquires implicitly. SAM3 therefore offers an external source for the discretization, and with it a way to test whether a cleaner, object-aligned partition improves depth.

Foundation features for depth. A parallel line of work treats frozen self-supervised or depth-pretrained ViT features as depth backbones, among them DINOv2 [6] and Depth Anything [8]. Frozen features clearly suffice to drive depth, but what remains unclear is which pretraining matters. The features used here are segmentation-pretrained, and the results below point to that choice as the binding constraint. Confirming it would require a depth-relevant backbone behind the same head, which is left to future work.

Slot- and object-centric representations. Slot Attention and related object-centric methods decompose a scene into latent slots, typically under reconstruction or contrastive objectives that preserve object identity. iDisc’s IDRs are slots of the same kind, but trained by a depth objective alone, which, as it turns out, erases the grounded object structure imposed on the slots. That outcome is a caution for object-centric bottlenecks in dense geometric regression.

3. Method

3.1. Background: iDisc

iDisc [7] computes depth in two stages. First, a backbone and a multi-scale deformable-attention (MSDA) pixel decoder produce a feature pyramid $\{F_\ell\}$. An Adaptive Feature Partitioning (AFP) module then condenses those features, through iterated slot attention, into a small set of *Internal Discrete Representations* (IDRs) $Q \in \mathbb{R}^{N \times d}$ with N on the order of tens, a learned and ungrounded scene partition. The Internal Scene Discretization (ISD) head reconstructs depth from pixels features, which act as the attention queries (sinks), and the IDRs, which are mapped to keys and values,

$$\hat{d} = \text{ISD}(\{F_\ell\}, Q), \quad Q = \text{AFP}(\{F_\ell\}). \quad (1)$$

This paper measures the relative importance of these two information paths.

3.2. SAM3 as a controllable discretization

We now describe the combining of iDisc with SAM3. The SAM3 backbone provides both the dense feature pyramid, as well as K object slots (decoder queries and their masks and scores). We use those outputs in the two information paths described in Sec. 3.1, defining three orthogonal axes in our experiments: the pixel source, the IDR source and the optional unfreezing of SAM3 (Fig. 2). iDisc’s ISD and depth head are kept intact, and only the sources change, so that any change in depth can be attributed to them.

Pixel source (the feature path). The ISD features come either from iDisc’s multi-scale deformable-attention pixel decoder over SAM3’s FPN (MSDA), or from SAM3’s own text-fused encoder memory lifted to a pyramid by a light learned neck.

IDR source (the partition path). This is the axis under test. The compared IDR sources are:

- **Linear**, where K SAM3 queries are projected directly into IDRs. This is the simplest swap.
- **Attention adapter**, in which the SAM3 queries self-attend and cross-attend over the feature map into K refined IDRs. This is a standard learned refinement, the

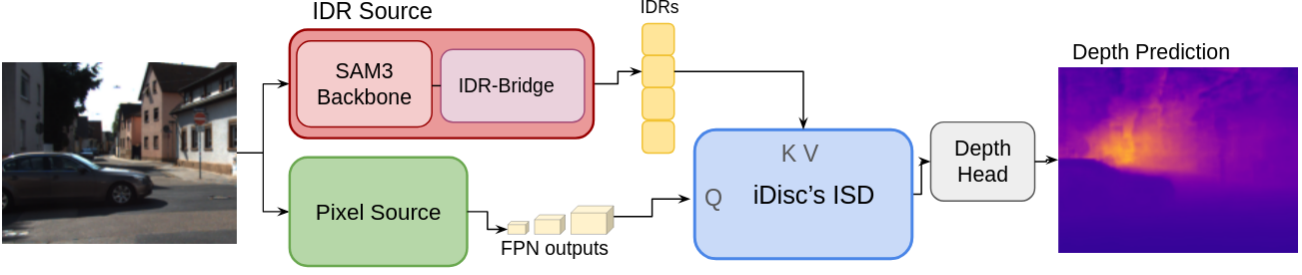


Figure 2. **Architecture.** A frozen SAM3 backbone feeds iDisc’s depth head. Two axes are swappable: the *pixel source* (FPN or fused memory) that drives the ISD features, and the *IDR source* that drives the discretization (SAM3 backbone provides queries or masks, from which IDRs are constructed via linear projection or an attention refinement in the IDR-Bridge).

conventional design for giving the partition path a fair chance (Sec. 3.3).

- **Mask-grounded linear**, in which the features inside each SAM3 object mask are mask-weighted average-pooled into one center per object, then projected with the same per-resolution Linear as the linear variant. This isolates the partition’s *source*: it is the linear head with its input swapped from raw queries to mask-grounded centers, on equal footing (same projection, head, and training), so any difference reflects the source rather than the capacity.

Unfreezing. The SAM3 post-trunk neck, encoder and decoder are optionally fine-tuned at low learning rate to test whether moving the *features* (not the partition) changes depth.

3.3. A standard learned refinement (the adapter)

The linear projection treats each SAM3 query in isolation. The adapter instead lets the K object tokens reorganize against the image before they become IDRs, at each of the L ISD resolutions independently. The block follows a conventional, DETR-style [1] query-refinement layer. Writing $T^{(0)} \in \mathbb{R}^{K \times d}$ for the SAM3 tokens and F_ℓ for the ℓ -th feature map, each of D blocks applies token self-attention, token→pixel cross-attention, and an MLP,

$$T \leftarrow T + \text{SelfAttn}(T), \quad (2)$$

$$T \leftarrow T + \text{CrossAttn}(T, F_\ell + \text{PE}(F_\ell)), \quad (3)$$

$$T \leftarrow T + \text{MLP}(T), \quad (4)$$

producing one IDR set $Q_\ell \in \mathbb{R}^{K \times d}$ per resolution ($d=256$, $D=2$). The cross-attention keys are the *full*-resolution feature map, with no pooling. Because the module trains from scratch and the rest of the head is unchanged, any difference from the linear baseline is due to it alone. It is not a novel module, and it partly repeats SAM3’s own frozen decoder, which has already refined these tokens against image features. Its role is only to give the partition path a

fair, adequately sized learned attempt, not to contribute one. No more elaborate generator is pursued, since the source-invariance result (Sec. 5.1) already ties the bare linear projection to the mask-grounded source, leaving no room for a fancier one to separate from them.

3.4. Temporal-consistency loss for video

For the video setting (Sec. 5.4), a temporal-consistency loss is added to the released iDisc-R101, with a frozen SAM3 video tracker serving as a side branch. At weight $\lambda = 0$ the model reduces to iDisc. The tracker assigns each object a consistent instance id across a clip. Writing $\bar{d}_k^t$ for object k ’s mask-pooled depth in frame t , the loss penalizes the second temporal difference of its *log* depth,

$$\mathcal{L}_{\text{tmp}} = \sum_k \left\| \log \bar{d}_k^{t-1} - 2 \log \bar{d}_k^t + \log \bar{d}_k^{t+1} \right\|, \quad (5)$$

with total loss $\mathcal{L}_{\text{depth}} + \lambda \mathcal{L}_{\text{tmp}}$. The log-Laplacian vanishes for locally linear motion, so the term penalizes frame-to-frame jitter rather than legitimate ego or object motion, and the logarithm makes it scale-invariant. Sec. 5.4 evaluates this loss.

4. Experimental Setup

Data. We evaluate our method on a KITTI [2] Eigen split with 23,158 training images and 652 testing images, with single-image input and sparse projected-LiDAR ground truth (capped at 80 m). The temporal study additionally samples non-overlapping 4-frame clips from the raw KITTI sequences (61 drives).

Training. The ~ 12 – 49 M IDR-Bridge is trained from random initialization. Additionally, when finetuning SAM3, only the post-trunk neck, encoder and decoder are trained. SAM3 ViT trunk and text encoder (~ 840 M params) always stay frozen. We use an AdamW [4] optimizer with a OneCycleLR schedule, with maximum learning rate 5×10^{-5} for the bridge and 1×10^{-5} for SAM3

IDR source	steps	AbsRel ↓	δ_1 ↑	RMSE ↓
ResNet-101 (<i>baseline</i>)	45k	0.0600	0.9638	2.362
SAM3 linear (queries)	5k	0.0691	0.9546	2.470
SAM3 linear (queries)	15k	0.0618	0.9643	2.347
SAM3 adapter (attention)	5k	0.0685	0.9548	2.459
SAM3 adapter (attention)	15k	0.0607	0.9652	2.310
SAM3 mask linear (masks)	5k	0.0689	0.9549	2.494
SAM3 mask linear (masks)	15k	0.0619	0.9638	2.326

Table 1. **Depth accuracy is invariant to the partition’s source.** SAM3 rows use the memory pixel source with the post-trunk unfrozen; the baseline is the public checkpoint, re-evaluated in this pipeline. At the matched 5k budget, linear, adapter, and mask-grounded sources span 0.0006 AbsRel and 0.0003 δ_1 , all trailing the baseline, showing that the source of the partition does not change accuracy. At 15k the same three sources stay within 0.0012 of one another and reach the baseline (the adapter passing it on δ_1 and RMSE), so the longer budget is the one closing the gap, not the source.

finetuning. For the depth loss $\mathcal{L}_{\text{depth}}$, we use the same $\text{SI}_{\log}$ loss as [7]. We run 5k or 15k iteration steps with effective batch size 2 (micro-batch 1 with gradient accumulation). Images are letterboxed into 1008^2 input, with similar augmentations as in [7].

Metrics. We use standard absolute relative error (AbsRel), root mean square error (RMSE) and the percentage of inlier pixels with threshold 1.25 (δ_1) on the Eigen test set. The partition analyses additionally report the IDR-ablation deltas (Sec. 5.2) and the R^2 of a linear fit of pixel depth to per-region mean depth (Sec. 5.3).

5. Results

5.1. Depth is invariant to the partition’s source

By fixing the pixel source and the training budget, we can observe the influence the IDR source has on the depth estimation. Linear and adapter reach AbsRel of 0.0691 and 0.0685 respectively, and mask-linear reaches 0.0689 (Tab. 1). Since the grounded mask-linear source shares the projection, head, and budget of the linear one, the comparison isolates the effect of grounding alone. However, that effect seems to be absent and the predictions are visually indistinguishable between sources (Fig. 3). Training to a longer budget leaves this unchanged, as at 15k iterations the three sources move together to AbsRel 0.0618, 0.0607, and 0.0619, still within 0.0012 of one another. They close most of the AbsRel gap to the baseline and match it on δ_1 and RMSE, with the adapter surpassing it on both (0.9652 and 2.310). The added training, not the choice of source, drives this gain.

Head	pixel	off	zero (Δ)	swap (Δ)
SAM3 linear	mem	0.0691	0.185 (+0.116)	0.113 (+.043)
SAM3 adapter	mem	0.0685	0.287 (+0.219)	0.111 (+.043)
SAM3 linear	MSDA	0.0733	0.196 (+0.123)	0.094 (+.021)
ResNet-101	MSDA	0.0600	0.0668 (+.0068)	0.0612 (+.0012)

Table 2. **Internal-discretization ablation** (KITTI Eigen test). *zero* = IDRs $\rightarrow$ constant (total partition-path value); *swap* = a foreign scene’s IDRs on every frame (scene-specific value). On every SAM3 head the partition is heavily load-bearing and scene-specific, regardless of IDR source or pixel source; only the fully-trained ResNet-101 is near-idle. This refutes the hypothesis that the partition is bypassed: the invariance in Tab. 1 is interchangeability, not disuse. (Shuffle is a no-op: permutation-invariant attention; verified $0.05996 \approx \text{off}$.)

5.2. The partition is load-bearing, not bypassed

One interpretation of source-invariance (Sec. 5.1) is that the head disregards the partition. We evaluate that by disabling the IDR path in two ways: under *zero*, the IDRs are constant, which removes the path entirely, and under *swap*, a fixed foreign scene’s IDRs are pasted onto every frame, so a partition remains but it belongs to the wrong scene.

Every SAM3 head turns out to depend on it. Zeroing increases AbsRel several-fold (+0.116 for linear, +0.219 for the adapter), and even the wrong scene’s partition costs +0.02 to +0.04, far above the 0.0006 spread between IDR sources in Tab. 1. The original iDisc setup with ResNet-101 backbone is the exception, as its partition stays nearly idle (*zero* +0.0068, *swap* +0.0012). Being fully end-to-end and with a depth-rich pyramid, it can route depth around the bottleneck, whereas the small SAM3 heads on top of a frozen segmentation trunk must route depth through the discretization. The same pattern holds whether the feature path is the coarse fused memory or the full MSDA pyramid, ruling out resolution as the cause.

The partition is thus both used and scene-specific, since the swap degrades accuracy, yet neither its method of generation nor its grounding alters accuracy (Tab. 1). This suggests that, to the head, the discretization behaves as a generic, scene-specific container: any encoder that fills it with the current scene yields the same depth, and filling it with object-grounded content is no better than filling it with ungrounded content.

5.3. Depth supervision erases the partition

In an attempt to further explain source invariance, we examine whether depth supervision degrades object structure. We measure region/depth coherence, quantified as the per-image R^2 of ground-truth depth explained by per-region mean depth. Each pixel under a confident SAM3 mask ($\sigma > 0.5$) is assigned to its arg-max mask region and pre-

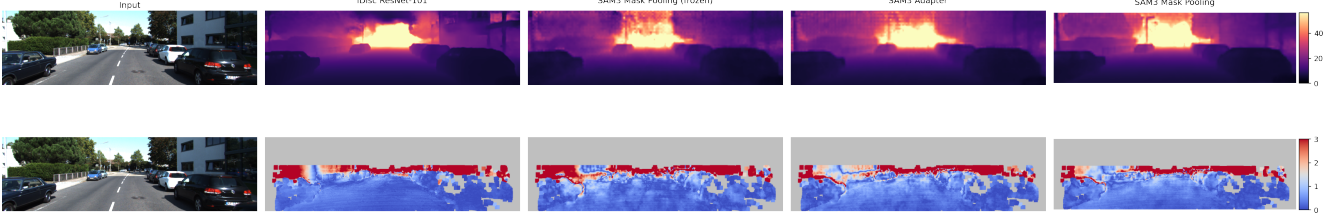


Figure 3. **Qualitative depth predictions (top) and error maps (bottom).** The iDisc ResNet-101 baseline and the SAM3 variants are visually near-identical, suggesting that the depth signal does not depend on the partition’s source.

SAM3 masks	mean R^2	median R^2	# active masks
frozen	0.75	0.76	~196
fine-tuned (end-to-end)	0.26	0.21	~147

Table 3. **Depth supervision erases grounding.** Region/depth coherence R^2 over the full 652-frame KITTI Eigen test set (mask confidence $\sigma > 0.5$). Fine-tuning the SAM3 post-trunk under the depth loss collapses the masks’ depth-coherence by two thirds and decreases the number of active masks.

dicted by that region’s mean depth, and the fit is scored over the 652-frame Eigen test set (Tab. 3 and Fig. 4).

Results suggest that frozen SAM3 masks are strongly depth-coherent, at mean R^2 0.75 and median 0.76. Fine-tuning the SAM3 post-trunk under the depth loss cuts this to mean R^2 0.26 and median 0.21, roughly a two-thirds drop, while the count of active masks falls from ~196 to ~147. Because the head reads the partition as a generic feature container rather than an object map, the objective is free to reshape it away from objects, and the grounding disappears. The object structure of the partition is a by-product of depth rather than its cause, since the partition stays in use (Sec. 5.2), only no longer through any correspondence to objects.

5.4. Video: temporal consistency

Video is the setting that motivated this work. Depth estimated independently per frame flickers over time, and tracking object tokens across frames could constrain each object’s depth to change smoothly. We test that mechanism as an auxiliary loss while keeping the depth model fixed. The predictor is the released single-image iDisc-R101, run independently on each frame. SAM3 enters only as a *video tracker* side branch, never as the depth backbone. The tracker takes a text prompt, which is propagated across the clip to assign each object a consistent instance id, and feeds those identities to the loss alone, not to the depth head. We sample non-overlapping 4-frame clips from the raw KITTI sequences (61 drives), and for every tracked instance penalize the second temporal difference of its mask-pooled log depth (Eq. (5)). The loss weight is swept over

λ (temporal loss)	AbsRel↓	flicker↓
0 (off)	0.0575	≈ 0.11
0.1	0.0605	≈ 0.11
0.3	0.0612	≈ 0.11

Table 4. **Temporal-consistency loss on KITTI clips.** AbsRel rises with the loss weight and the flicker measure stays within a ± 0.006 band of its $\lambda=0$ value, with no downward trend.

$\lambda \in \{0, 0.1, 0.3\}$, where $\lambda=0$ recovers plain iDisc. Each configuration is reported with two numbers, KITTI AbsRel and a per-instance flicker measure, the latter the magnitude of the temporal Laplacian on held-out clips.

The results suggest that the temporal stability does not improve (Tab. 4). As the loss weight grows, single-frame accuracy degrades monotonically, with AbsRel climbing from 0.0575 at $\lambda=0$ to 0.0612 at $\lambda=0.3$. The flicker measure, meanwhile, stays within ± 0.006 of its $\lambda=0$ value throughout the sweep, showing no downward trend. Two factors explain this. On single-frame KITTI, flicker is dominated by the frame-independent FPN prediction, which a per-instance penalty cannot reach. On top of that, the tracker recovers few confident instances per clip, leaving the loss with little to act on. The sweep is small, so the result is preliminary. Even so, it speaks to the regime that motivated the project, and it agrees with the erasure result of Sec. 5.3 that the object structure this approach needs has already been degraded by the depth objective.

6. Discussion

What the depth head uses. In Sec. 5.2, we showed how the partition is not omissible for SAM3 heads. What can be varied freely is how that partition is grouped. Sourcing it from SAM3 doesn’t change accuracy whatever the construction, so the source, quality, and grounding of the discretization are all irrelevant. A better partition therefore does not yield better depth on KITTI even though it is used. Whether the SAM3 heads’ reliance on the partition comes from the frozen segmentation trunk or from the small from-scratch head remains open, and is left to future work.

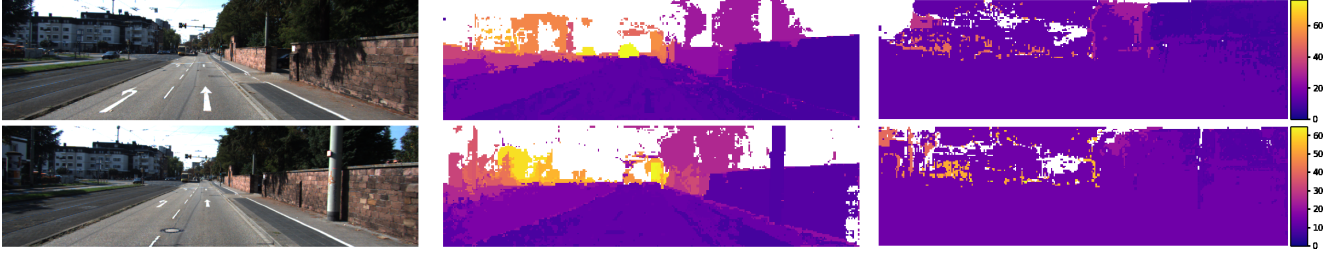


Figure 4. **Fine-tuning breaks the partition.** Two scenes (rows): RGB, frozen SAM3 masks (depth-coherent), fine-tuned SAM3 masks (mixed depths), color = per-region mean depth.

Implication for video. The project’s original goal was to damp temporal flicker by propagating tracked object tokens across frames. The erasure result (Sec. 5.3) shows why that is not viable in the present model. The depth objective destroys the very object structure (R^2 $0.75 \rightarrow 0.26$) that tracking would rely on, and the temporal-consistency study (Sec. 5.4) agrees, with flicker holding steady as accuracy worsens. For this family of models, temporal stability must come from the features, or from an explicit temporal model, rather than from an object-token bottleneck.

Beyond KITTI. KITTI is a hard case for this task. Its ground truth is sparse projected LiDAR, its aggregate metric is dominated by road and far field, and few confident SAM3 objects appear per frame, all of which make a grounded object partition least likely to help. The natural worry is whether the result survives in a denser, more object-centric regime, which a comparison on NYU (dense ground truth, object-rich indoor scenes) addresses. Following the matched protocol (memory pixel source, post-trunk unfreeze, 5k iterations), the linear and mask-grounded sources are fine-tuned on NYU-Depth V2 and evaluated on the official 654-image test split. The spread is again negligible: the mask-grounded source reaches AbsRel 0.1208 and the linear projection 0.1212, a gap of 0.0004 that, as on KITTI, sits far below the feature and ablation effects of Sec. 5.2. Both trail the end-to-end ResNet-101 baseline, re-evaluated in this pipeline at 0.1146, mirroring KITTI. The source of the partition is therefore inert indoors as well, a grounded population proving no more accurate than an ungrounded one. Training here uses a public HuggingFace NYU export rather than the original split, which shifts absolute accuracy but not the matched comparison, since both variants share it.

Limitations. Running a frozen DINOv2 or Depth-Anything backbone through the same head would test whether the limit is SAM3’s segmentation features in particular rather than frozen features in general. That test, together with the question of why the SAM3 heads rely on the partition while ResNet-101 does not, is left to future work.

The reverse direction was also explored, namely grounding iDisc’s own AFP partition toward SAM3 with auxiliary affinity and boundary losses. Consistent with the source-invariance result, these did not lower AbsRel in preliminary runs, and a fuller study of loss design, weighting, and denser supervision remains for future work.

7. Contributions, Work Distribution, and Code

Novelty. New for iDisc and SAM3, to our knowledge: (i) depth accuracy is invariant to the partition’s source and grounding (linear, attention, and grounded-mask sources all tie); (ii) the IDR zero/swap ablation shows this is fungibility rather than disuse, since the partition path is load-bearing and scene-specific while its grouping is irrelevant; (iii) a depth objective erases grounded object structure, which precludes object-token tracking for temporal stability in this model. The result is negative but mechanistic.

Code. Source released at <https://github.com/Fersoil/idisc>. *Ours*: the swappable IDR/pixel-source bridge, the attention adapter and mask linear, the SAM3 unfreezing path, the temporal-consistency loss, and the ablation / region- R^2 tooling. *External*: iDisc [7] (model, training loop, released checkpoints, prebuilt MSDeformAttn op) and SAM3 [5] (backbone + weights); PyTorch 2.12 (CUDA 12.8), Hydra, OmegaConf, timm, einops, OpenCV.

Work distribution.

- M. Shpir — SAM3 $\leftrightarrow$ iDisc integration and the swappable IDR/pixel-source bridge, the mask linear source, and the IDR-ablation;
- T. Kwieciński — the SAM3 video-tracker branch and the temporal-consistency loss, the IDR visualizations, the memory/bf16 training path, W&B logging, and the install and caching setup;
- L. Nidogon — the sequential-data baselines and the fine-tuned baseline implementation, the NYU pipeline;
- S. Lazarusic — the sequential-data validation and evaluation, the visualizations, the region- R^2 analysis and writeup.

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